OLYMPIC INSIGHTS

Using Data Analytics

An End-to-End Power BI Analytics Project

Presented by

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Power BI · Power Query · DAX · Star Schema Data Modeling

Dataset: 6 Historical Olympic CSV Files · 1896–2022 · 315,626 Event Records

Table of Contents

What this presentation covers

Project Overview & Business Problem

Why this project was built and what it delivers

Business Problem

The International Olympic Committee (IOC) required advanced insights into athlete performance, country-wise medal trends, and sport-specific dominance across Olympic history — to support strategic planning and athletic development decisions.

What This Project Delivers

- 6-page interactive Power BI dashboard covering 126 years of Olympic history
- 315,626 event records profiled, cleaned, and modeled into a star schema
- 7 validated model relationships confirmed against live data
- Custom DAX measures for all KPIs — tested through visual cross-checks
- Drill-through pages for country-level and sport-level deep dives

Tools & Technologies

Power BI Desktop

Interactive dashboards, data model, DAX measures, slicers, drill-through pages

Power Query (M)

Data loading, cleaning, type conversion, custom columns, duplicate removal

DAX

KPI measures, calculated columns, RANKX, TOPN, MAXX, SELECTEDVALUE, DIVIDE

Star Schema

3 fact tables, 3 dimensions, 7 relationships, single-direction cross-filtering

Source Data

6 CSV files — 1896 to 2022 — verified against live model during build

Dataset Overview

6 source files · 480,000+ combined records · Olympic history 1896–2022

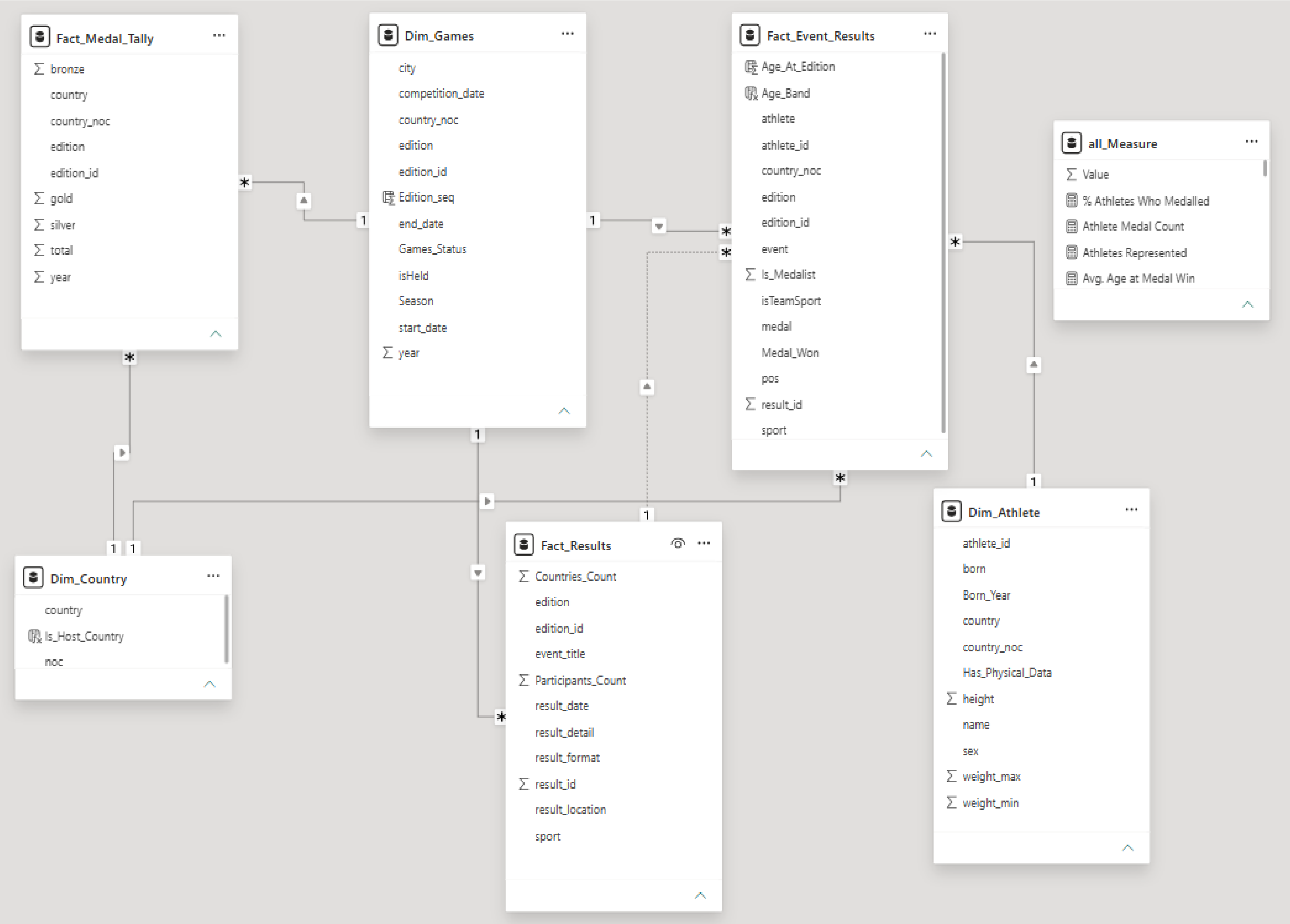
Table Name	Rows	Grain	Description
Dim_Athlete	155,861	1 row / athlete	Athlete biography — name, sex, born year, height, weight, country NOC
Fact_Event_Results	315,626	1 row / athlete–event–edition	Core transactional fact — individual medal results, sport, event per athlete
Fact_Medal_Tally	1,807	1 row / country–edition	Pre-aggregated medal totals — gold, silver, bronze per country per Games
Fact_Results	7,394	1 row / event–edition	Event metadata — participant count, result format, result detail
Dim_Country	235	1 row / NOC code	Country reference — NOC code and full country name
Dim_Games	64	1 row / edition	Games reference — year, city, season, host NOC, status

Critical Design Rule

Country-level medal totals always source from Fact_Medal_Tally (pre-aggregated, team-safe). Athlete-level and sport-level analysis always sources from Fact_Event_Results (individual grain). These two tables must never be swapped — this distinction drives the correct DAX design across all six dashboard pages.

Data Model — Validated Star Schema

7 relationships confirmed · 3 fact tables · 3 dimensions · Import mode



Model Decisions

edition_id keys

All Dim_Games relationships use numeric edition_id — not the edition text column.

No Calendar table

Dim_Games is the time dimension, extended with Edition_Sequence (RANKX ordinal).

Dim_Athlete→Dim_Country removed

Created ambiguous filter path. Fact_Event_Results→Dim_Country is the correct path.

Fact→Fact Inactive

Fact_Event_Results ↔ Fact_Results on result_id — kept Inactive to avoid triangle loop.

Single-direction filters

All cross-filters: dimension → fact only.

Global Medal Overview — Summer & Winter Olympics, 1896–2022

A century-plus of the Olympic Games — medals, nations, and events at a glance

Total Medals Won	Countries Participated	Total Editions Held	Total Events
20,412	230	53	963

FILTER

Season

- Summer
- Winter

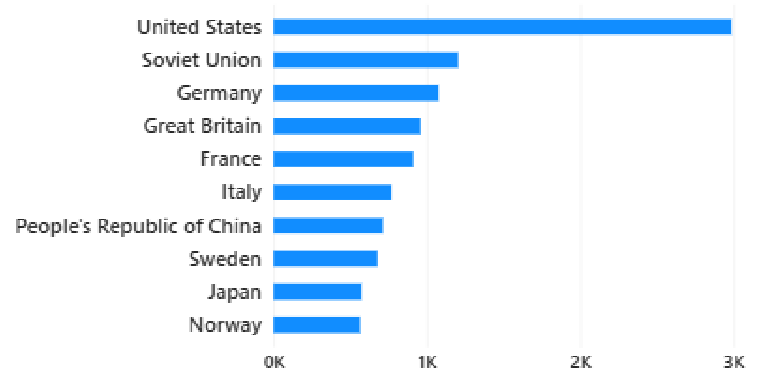
Games_Status

Held

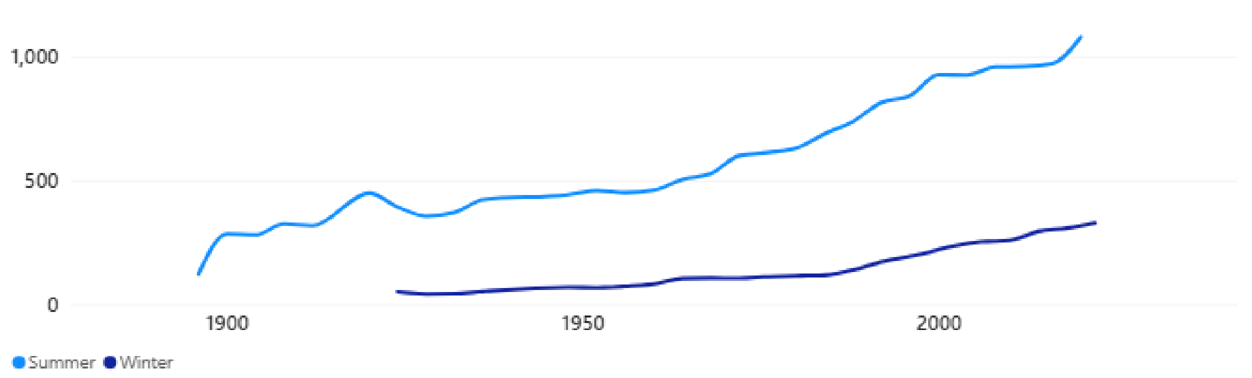
year

1896 2022

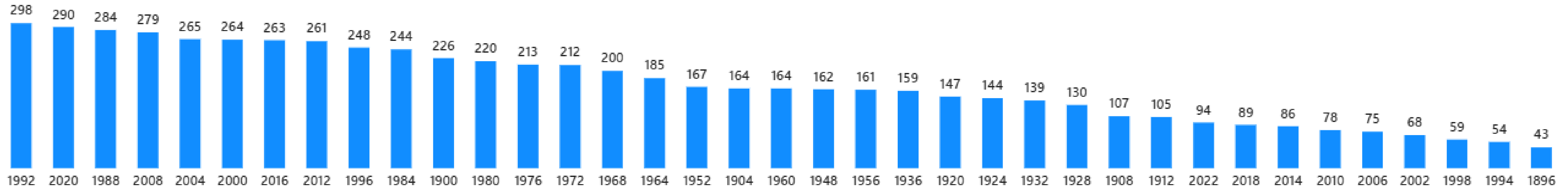
Top 10 Countries by Total Medals



Medal Trends Over Time



Events Held by Year



Page 1 — Global Medal Overview

DAX Measures

DAX Measures & Analytical Logic — Page 1 — Global Medal Overview

Total Medals Won

```
Total Medals Won = SUM ( Fact_Medal_Tally[total] )
```

→ *Sums the pre-aggregated total column — sources from Fact_Medal_Tally to correctly avoid team-sport double-counting.*

Total Countries Participated

```
Total Countries Participated =
 (DISTINCTCOUNT ( Fact_Event_Results[country_noc] )
```

→ *Counts all nations that sent athletes. Explicitly excludes "IFR" — a non-national placeholder code.*

Total Editions Held

```
Total Editions Held =
  CALCULATE (
    DISTINCTCOUNT ( Dim_Games[edition_id] ),
    Dim_Games[Games_Status] = "Held")
```

→ *Counts only editions with Games_Status = "Held" — excludes the 5 war-cancelled and future editions.*

Total Events Held

```
Total Events Held =(DISTINCTCOUNT ( Fact_Event_Results[event] )
```

→ *Distinct event names across all held editions only.*

Page 1 Dynamic Title

```
Page1_Dynamic_Title =
VAR SelectedSeason =
           SELECTEDVALUE (Dim_Games[Season], "Summer & Winter")
VAR MinYear =
           CALCULATE (MIN (Dim_Games[year]), Dim_Games[Games_Status] = "Held")
VAR MaxYear =
           CALCULATE (MAX (Dim_Games[year]), Dim_Games[Games_Status] = "Held")
RETURN
     "Glowbal Medal Overview - " & SelectedSeason & "Olympic, "& MinYear & "-" & MaxYear
```

→ *Updates the page heading dynamically when Season slicer changes. SELECTEDVALUE returns fallback when multi-select.*

Page 1 — Global Medal Overview

Business Insights

Total Medals Won	Countries Participated	Editions Held	Total Events
20,412	230	53	963

Business Insights

- USA leads all-time total medals by a clear margin — Soviet Union, Germany, Great Britain, and France follow. The top 10 nations account for a disproportionate share of 20,412 total medals awarded across 126 years.
- Summer Games medals have grown steadily from 1896 to 2022. Winter Games medals also grew but remain significantly smaller in scale — the two distinct trend lines are clearly visible in the Medal Trends Over Time chart.
- Events Held by Year confirms a 6.9× expansion of the Olympic programme: from just 43 events in 1896 to 298 in 1992 — the highest edition in this dataset.
- The Games_Status slicer defaults to "Held" — correctly excluding the five war-cancelled editions (1916, 1940×2, 1944×2) from all KPIs and trend lines.
- Top 10 Nations: United States · Soviet Union · Germany · Great Britain · France · Italy · China · Sweden · Japan · Norway

Slicers: Season (Summer / Winter) · Games_Status (default: Held) · Year Range: 1896–2022

Country Insights — United States

Medal efficiency, sport strengths, and year-over-year performance by nation

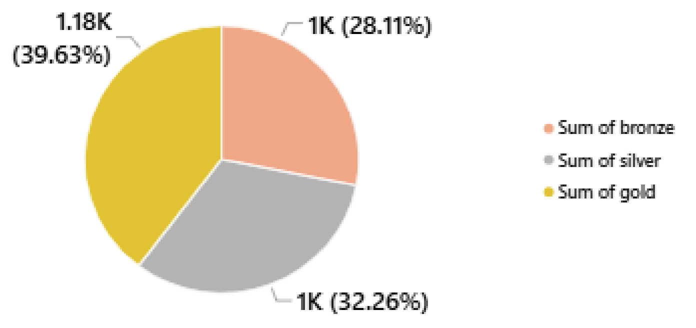
Athletes Represented	Total Medals Won	Medal Efficiency (per 100 Athletes)	Best Performing Sport
11,812	2,985	25.27	Athletics

FILTER

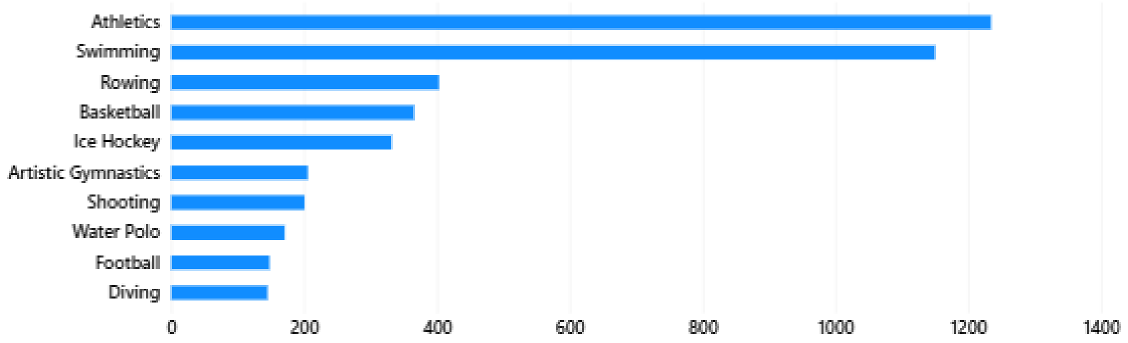
Games Season
- Summer
- Winter

country
- United States

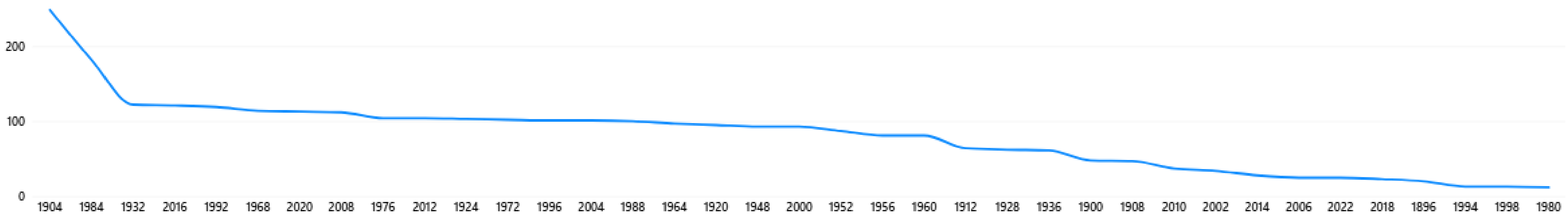
Medal Split



Top Sports by Medal Count



Medal Count By Year



Page 2 — Country Performance Insights

DAX Measures

DAX Measures & Analytical Logic — Page 2 — Country Performance Insights

Athletes Represented

```
Athletes Represented = DISTINCTCOUNT ( Fact_Event_Results[athlete_id] )
```

→ Counts distinct athletes who competed for the selected country across all editions.

Total Medals Won

```
Total Medals Won = SUM ( Fact_Medal_Tally[total] )
```

→ Country-level medal total sourced from Fact_Medal_Tally — the correct, team-safe, pre-aggregated table.

Medal Efficiency

```
Medal Efficiency =
  DIVIDE ( [Total Medals Won] * 100, [Total Athletes Represented], 0 )
```

→ Medals per 100 athletes. DIVIDE with 0 fallback safely handles countries with zero medals.

Best Performing Sport

```
Best Performing Sport =
  VAR SportMedals = ADDCOLUMNS (
    VALUES ( Fact_Event_Results[sport] ),
    "MedalCount", CALCULATE (
      COUNTROWS ( Fact_Event_Results ),
      Fact_Event_Results[medal] <> BLANK() )
)
  VAR TopSport =
      TOPN ( 1, SportMedals, [MedalCount], DESC, Fact_Event_Results[sport], ASC )
  RETURN
      CONCATENATEX ( TopSport, Fact_Event_Results[sport], ", " )
```

→ Returns top sport by medal appearances. Handles ties via CONCATENATEX — shows both names instead of returning blank.

Medal Appearances by Sport

```
Medal Appearances by Sport =
  CALCULATE (
    COUNTROWS ( Fact_Event_Results ),
    Fact_Event_Results[medal] <> BLANK())
```

→ Medal-winning rows per sport from Fact_Event_Results — the only table with a sport column.

Page 2 — Country Performance Insights

Business Insights — United States

Athletes	Total Medals	Efficiency	Best Sport
11,812	**2,985**	**25.27**	**Athletics**

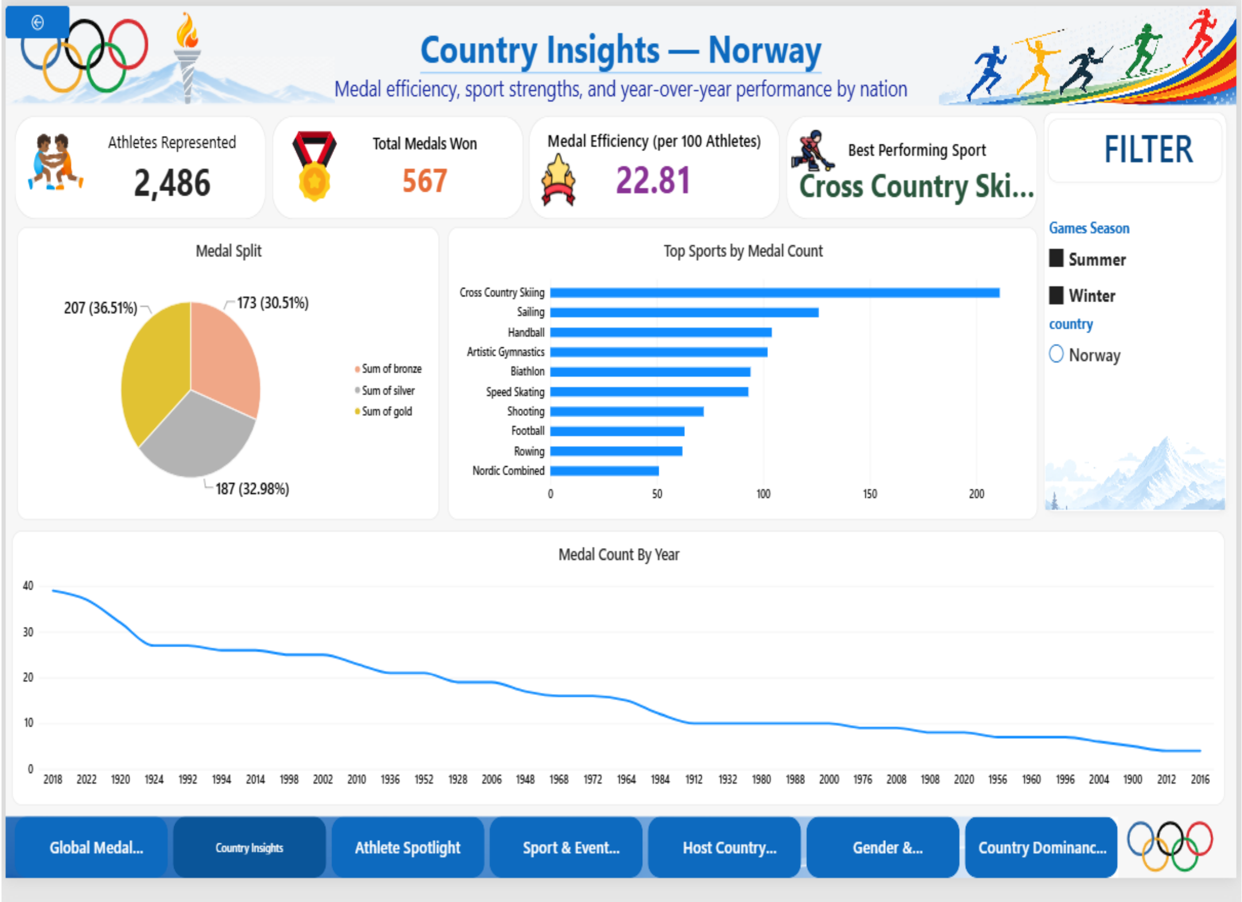
Business Insights

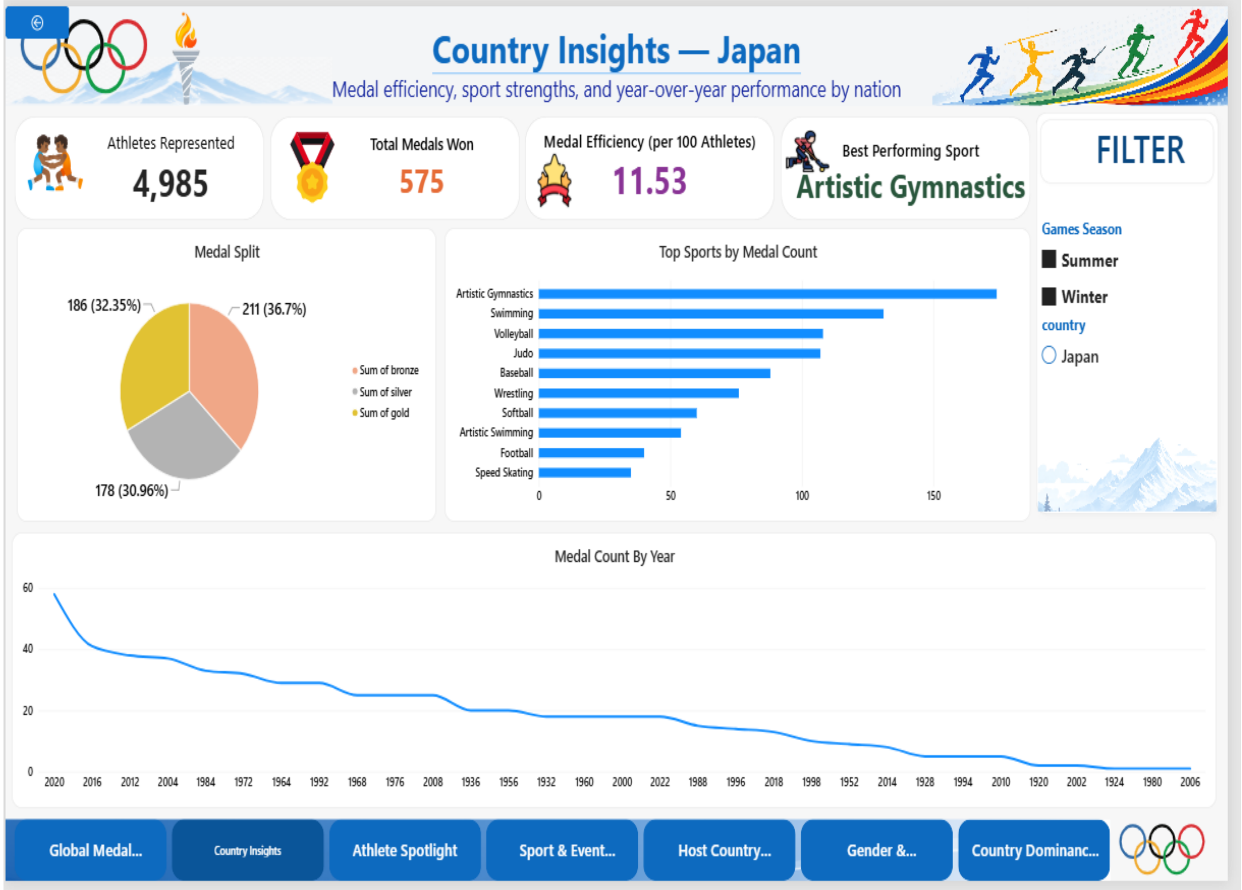
- USA — 11,812 athletes, 2,985 medals, efficiency 25.27 per 100. Highest medal total of all countries in the dataset. Gold at 39.63% — nearly 4 in 10 USA medals are gold.
- Athletics leads the sport breakdown by a large margin (1,400+ medal appearances), followed by Swimming and Rowing — consistent with USA–Athletics being the most dominant country-sport pairing overall.
- Medal split: Gold 39.63% · Silver 28.11% · Bronze 32.26%. Gold proportion significantly higher than silver — a programme structured to win rather than simply place.
- Medal Count by Year shows early editions produced the highest per-edition totals. The declining trend reflects growth in total available medals shared across more competing nations, not a decline in USA performance.
- Medal Efficiency of 25.27 confirms the USA as one of the most productive Olympic programmes relative to delegation size — competitive in both volume and efficiency.

Slicers: Dim_Country[country] (single-select) · Season · Drill-through entry from Page 1 Top 10 chart

Country Insights — Norway & Japan

Page 2 under different country selections





2,486 athletes | 567 medals | Efficiency: 22.81 per 100 | Best Sport: Cross Country Skiing
Gold 30.5% · Silver 33.0% · Bronze 36.5%

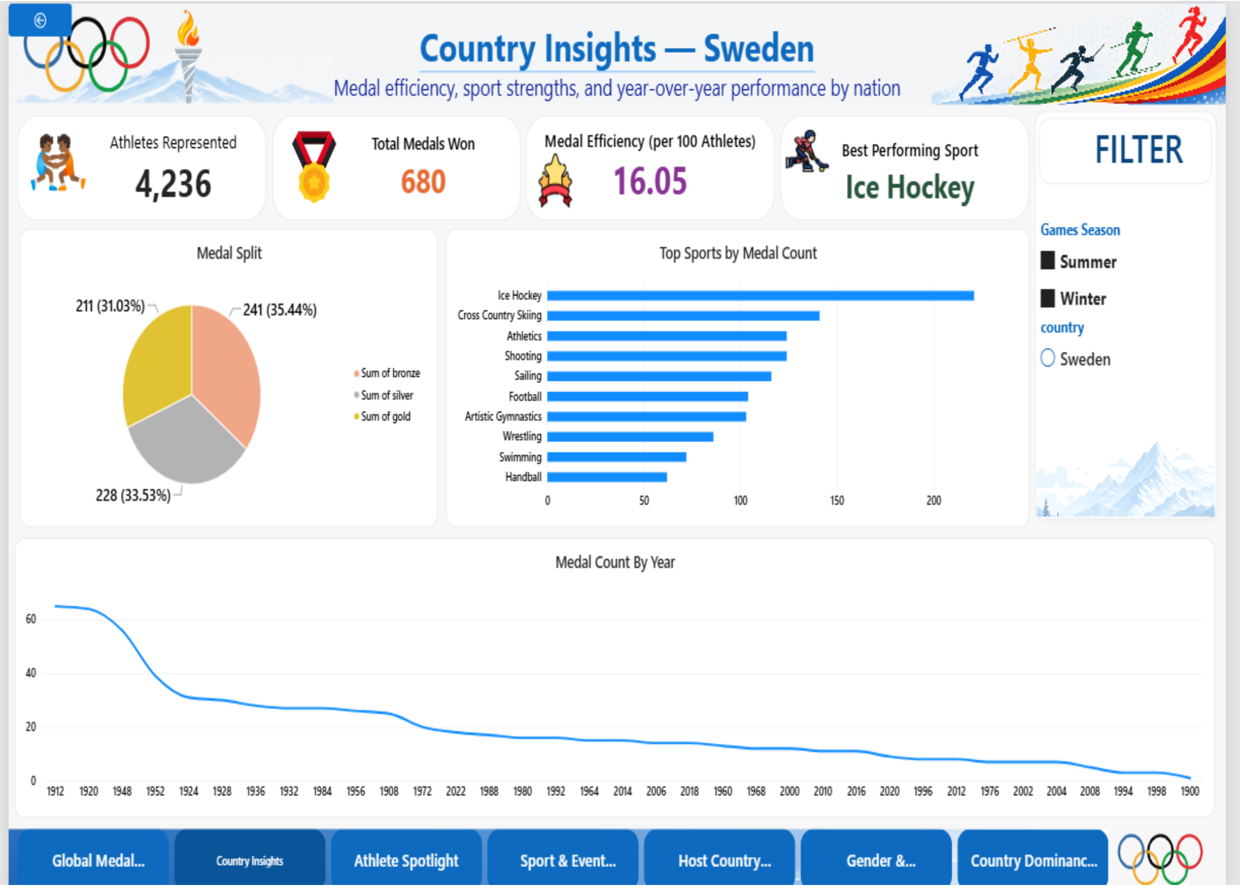
Near-balanced medal split indicates depth across all podium positions. Cross Country Skiing dominates the sport breakdown — a direct reflection of Winter Olympic specialisation. Norway achieves high efficiency with a relatively small delegation compared to other top nations.

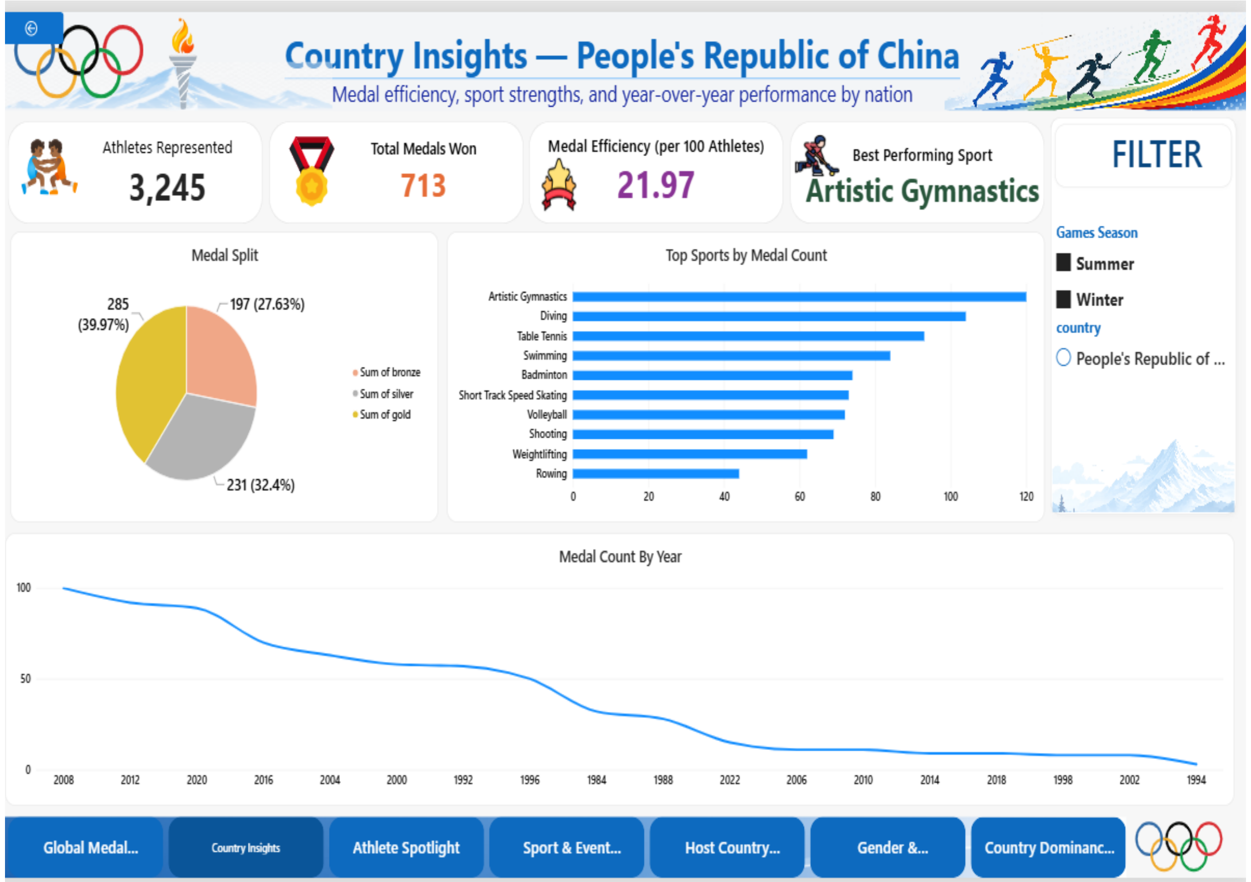
4,985 athletes | 575 medals | Efficiency: 11.53 per 100 | Best Sport: Artistic Gymnastics
Gold 36.7% · Silver 31.0% · Bronze 32.3%

Despite fielding twice the athletes of Norway, efficiency is roughly half — confirming that larger delegations do not proportionally increase medal returns. Gold is disproportionately higher than bronze, indicating a programme structured to target wins over general podium placements. Top sports: Artistic Gymnastics, Swimming, Volleyball.

Country Insights — Sweden & China

Page 2 under different country selections





4,236 athletes | 680 medals | Efficiency: 16.05 per 100 | Best Sport: Ice Hockey
Gold 35.4% · Silver 33.5% · Bronze 31.0%

Ice Hockey leads Sweden's sport breakdown by a clear margin — 200+ medal appearances, significantly ahead of the second-ranked sport. Gold proportion at 35.4% is the highest of all four countries shown. Sweden's programme spans both Summer and Winter Games effectively, making it one of the more versatile national programmes in the dataset.

3,245 athletes | 713 medals | Efficiency: 21.97 per 100 | Best Sport: Artistic Gymnastics
Gold 39.97% · Silver 32.4% · Bronze 27.6%

Highest medal total and second-highest efficiency of the four countries. Gold at 39.97% — the highest gold proportion in the dataset — reflects a programme structured around winning specifically. Top sports: Artistic Gymnastics, Diving, Table Tennis, Swimming, Badminton — all precision and technical disciplines.

Athlete Spotlight

Career longevity, top performers, and demographics of medal winners

Total Athletes	% Athletes Who Medalled	Most Olympic Appearances	Avg. Age at Medal Win
1,55,861	20.26%	10	26.36

FILTER

Games Season

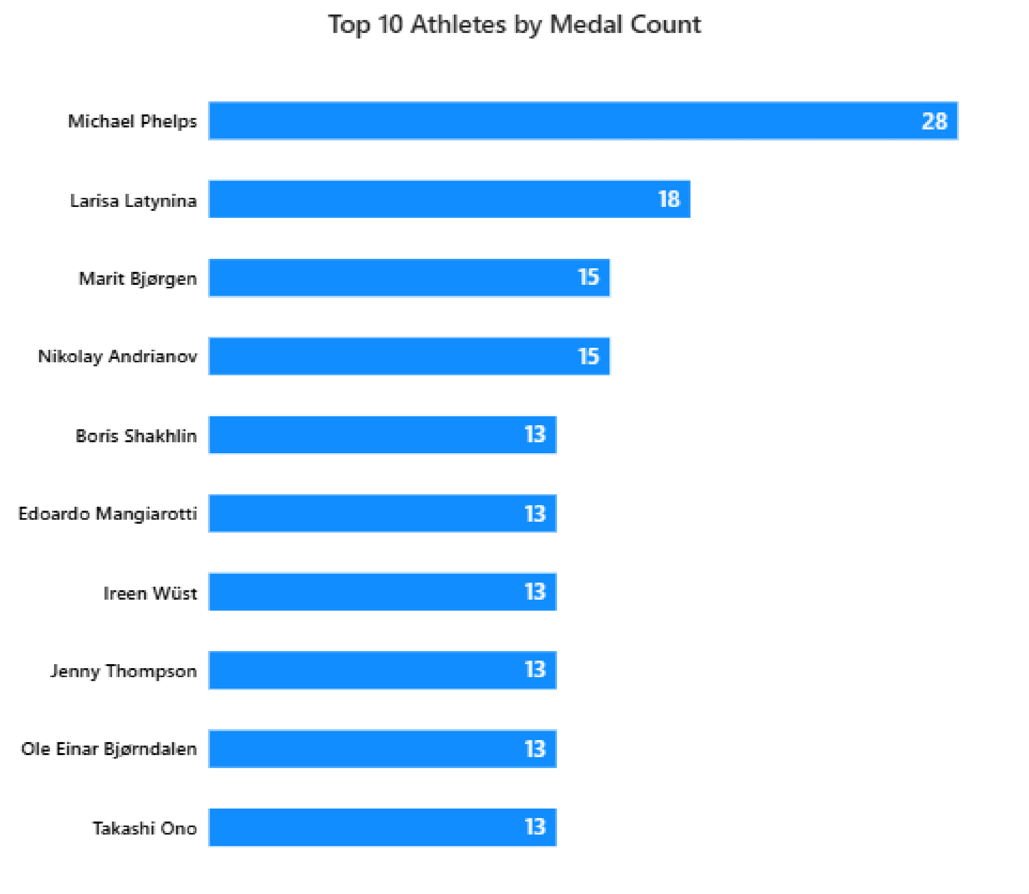
Multiple selections

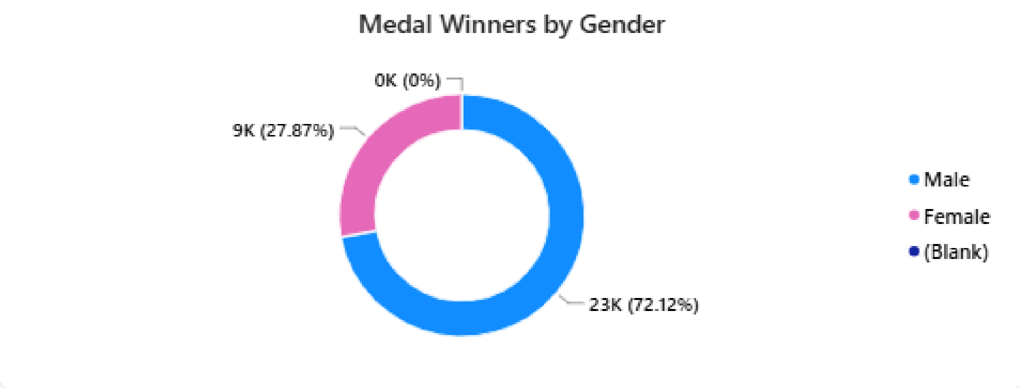
sex

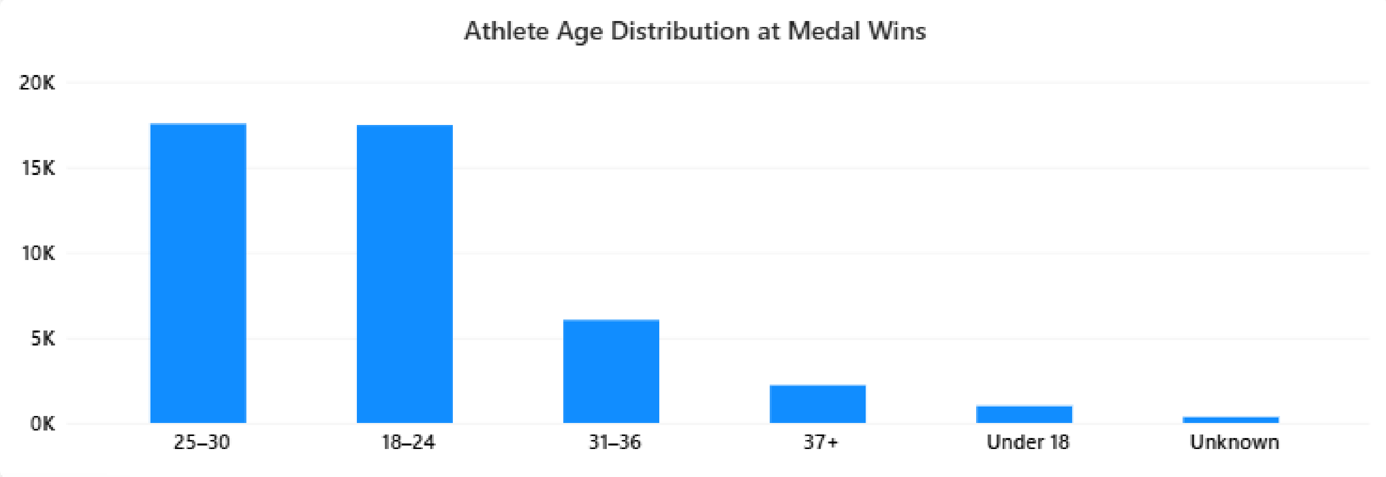
- Female
- Male

year

1896 2022







Global Medal... | Country Insights | Athlete Spotlight | Athlete Spotlight | Host Country... | Gender &... | Country Dominanc...

Page 3 — Athlete Spotlight

DAX Measures

DAX Measures & Analytical Logic — Page 3 — Athlete Spotlight

Total Athletes

```
Total Athletes = DISTINCTCOUNT ( Dim_Athlete[athlete_id] )
```

Medal-Winning Athletes & %

```
Medal-Winning Athletes =
  CALCULATE (
    DISTINCTCOUNT ( Fact_Event_Results[athlete_id] ),
    Fact_Event_Results[medal] <> BLANK() )

% Athletes Who Medalled =
  DIVIDE ( [Medal-Winning Athletes], [Total Athletes], 0 )
```

Most Olympic Appearances

```
Most Olympic Appearances =
  MAXX (
    VALUES ( Fact_Event_Results[athlete_id] ),
    CALCULATE ( DISTINCTCOUNT ( Fact_Event_Results[edition_id] ) ))
```

→ Maximum distinct editions per athlete. Inner CALCULATE is essential for context transition — without it, DISTINCTCOUNT returns the global total for every athlete.

Avg. Age at Medal Win

```
Avg. Age at Medal Win =
  CALCULATE (
    AVERAGE ( Fact_Event_Results[Age_At_Edition] ),
    Fact_Event_Results[medal] <> BLANK())
```

→ Average age across all medal-winning rows only — excludes non-medal participation.

Age_At_Edition (Power Query)

```
Age_At_Edition (Power Query custom column):
  GamesYear - Born_Year
  Applied bounds: if result < 10 or > 75 then null

Born_Year extraction:
  Digits = Text.Select(Text.Trim([born]), {"0".."9"})
  Candidate = Number.FromText(Text.End(Digits, 4))
  if Candidate >= 1850 and <= CurrentYear then Candidate else null
```

→ Born_Year built in Power Query using digit extraction with 1850–present range validation. Bounds 10–75 on Age_At_Edition convert outliers to blank.

Page 3 — Athlete Spotlight

Business Insights

Total Athletes	% Medalled	Most Appearances	Avg Age at Win
1,55,861	20.26%	10	26.36 yrs

Business Insights

- Michael Phelps leads with 28 medals — the highest individual tally in the dataset. Larisa Latynina follows with 18. Swimming and Gymnastics dominate the top 10, as both sports offer multiple individual medal events per Games.
- Only 20.26% of 1,55,861 competing athletes ever won a medal — confirming the elite threshold of Olympic achievement. Over 79% of all competing athletes finished without a podium result.
- The 25–30 age band is the largest group at medal wins, followed closely by 18–24. Together these two age groups account for the vast majority of all medal-winning performances.
- Male medal winners: 72.12% · Female: 27.87%. This reflects the much longer history of male-only or male-dominant participation across the full 1896–2022 dataset.
- Top 10 Athletes: Phelps(28) · Latynina(18) · Bjørgen(15) · Andrianov(15) · Shakhlin(13) · Mangiarotti(13) · Wüst(13) · Thompson(13) · Bjørndalen(13) · Ono(13)

Slicers: Season · Dim_Athlete[sex] (Male / Female) · Year Range: 1896–2022

Sport & Event Analysis

Where medals are concentrated — sport growth and country dominance

Total Sports	Total Events	Events Added vs Previous Edition	Most Dominant Pairing
112	963	5	USA — Athletics

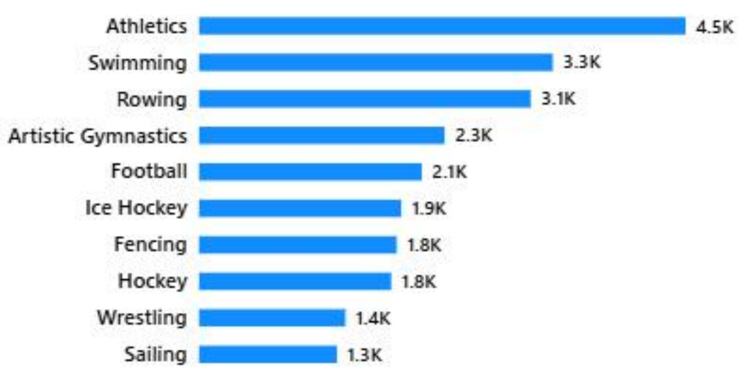
FILTER

Games Season: Multiple selections

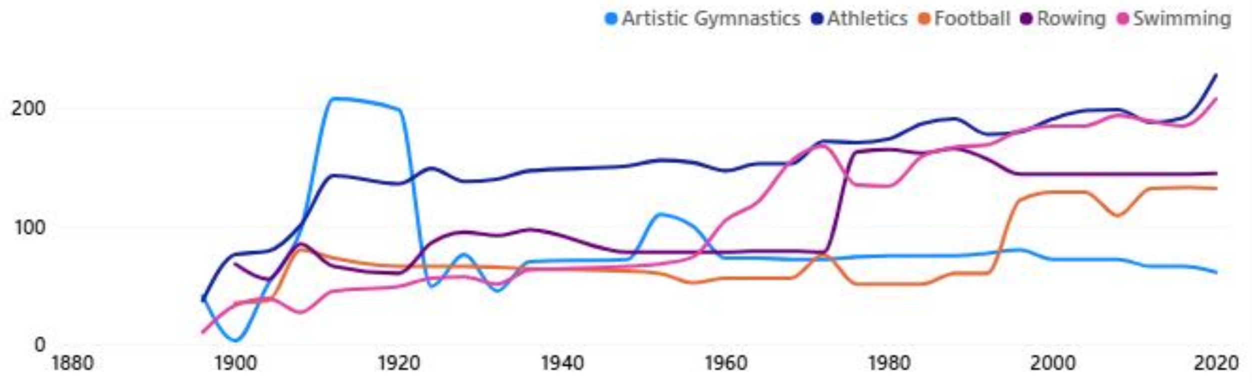
year: 1896 – 2022

sport
- Select all
- 3x3 Basketball
- Aeronautics
- Alpine Skiing
- Alpinism
- American Football
- Archery

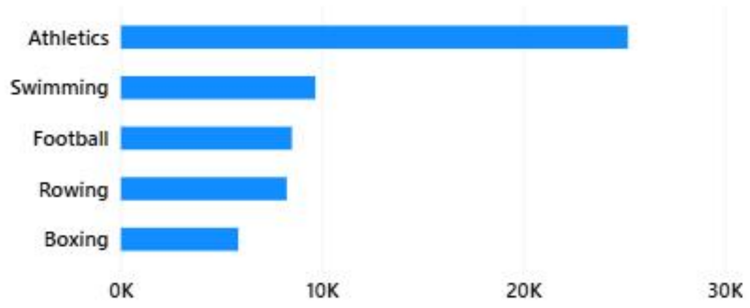
Top 10 Sports by Medals Awarded



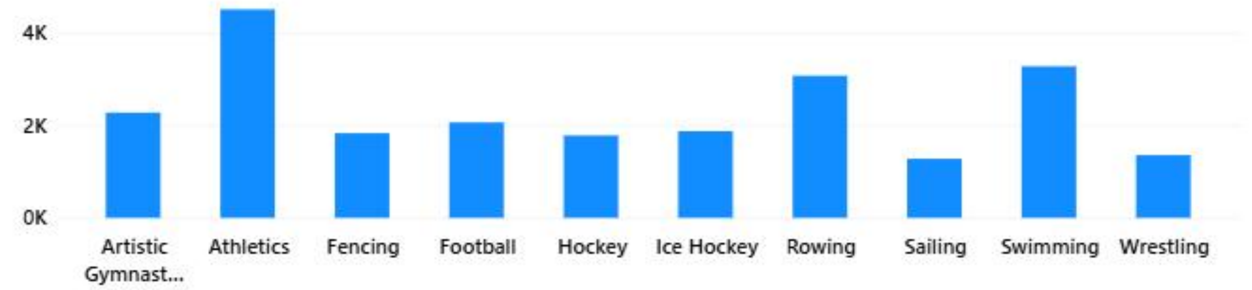
Sport-wise Medal Trends Over Time



Top Sports by Athlete Participation



Country Dominance by Sport



Page 4 — Sport & Event Analysis

DAX Measures

DAX Measures & Analytical Logic — Page 4 — Sport & Event Analysis

Total Sports & Events

```
Total Sports = DISTINCTCOUNT ( Fact_Event_Results[sport] )
```

```
Total Events = DISTINCTCOUNT ( Fact_Event_Results[event] )
```

Events in Latest Edition

```
Events in Latest Edition =
  VAR LatestHeldEdition =
      CALCULATE (MAX ( Dim_Games[Edition_id] ),Dim_Games[Games_Status] = "Held" )

  RETURN
      CALCULATE (
          DISTINCTCOUNT ( Fact_Event_Results[event] ),
          Fact_Event_Results[edition_id] = LatestHeldEdition )
```

→ Uses Edition_Sequence — not MAX(edition_id) which is non-chronological. Filtered to held editions only.

Events Added vs Previous Edition

```
Events Added vs Previous Edition =
  VAR CurrentSeq = CALCULATE ( MAX ( Dim_Games[Edition_Sequence] ), Dim_Games[Games_Status] = "Held" )
  VAR CurrentSeason =
        CALCULATE (
             SELECTEDVALUE ( Dim_Games[Season] ),
             FILTER ( ALL(Dim_Games), Dim_Games[Edition_Sequence] = CurrentSeq ) )
  VAR CurrentEvents =
        CALCULATE (
             DISTINCTCOUNT (Fact_Event_Results[event] ),
             Filter (ALL (Dim_Gaes )), Dim_Games[Edition_Sequence] = CurrentSeq ) )
  VAR PriorSeq =
        CALCULATE (
           MAX ( Dim_Games[Edition_Sequence] ),
           FILTER ( ALL(Dim_Games),
                    Dim_Games[Edition_Sequence] < CurrentSeq
                      && Dim_Games[Season] = CurrentSeason
                      && Dim_Games[Games_Status] = "Held" ) )
  VAR PriorEvents =
        CALCULATE (
           DISTINCTCOUNT (Fact_Event_Results[event]),
           Filter (ALL (Dim_Games), Dim_Games[Edition_Seq] = PriorSeq)

  RETURN   CurrentEvents - PriorEvents
```

→ Season-matched comparison — compares same-season editions only. Cross-season comparison produces misleading results.

Page 4 — Sport & Event Analysis

DAX Measures

DAX Measures & Analytical Logic — Page 4 — Sport & Event Analysis

Sport Athlete Participation

```
Sport Athlete Participation = DISTINCTCOUNT ( Fact_Event_Results[athlete_id] )
```

Edition_Sequence

```
Edition_Sequence =
  RANKX (FILTER ( ALL ( Dim_Games ), NOT ISBLANK ( Dim_Games[year] ) ),
    Dim_Games[year], , ASC, DENSE)
```

→ Ordinal sequence of editions by year. NOT ISBLANK excludes phantom blank rows from RANKX calculation.

Most Dominant Pairing

```
Most Dominant Pairing =
  VAR PairingTable =
      ADDCOLUMNS (
          SUMMARIZE (
             FILTER ( Fact_Event_Results, Fact_Event_Result[medal] <> BLANK() ),
             Fact_Event_Result[country_noc], Fact_Event_Result[sport] ),

          "MedalCount",
           CALCULATE (
               COUNTROWS ( Fact_Event_Results ), Fact_Event_Result[medal] <> BLANK() ) )

  VAR TopPairing =
      TOPN ( 1, PairingTable, [MedalCount], DESC,
           Fact_Event_Results[country_noc], ASC, Fact_Event_Results[sport], ASC )

  VAR TopCountry =
      MAXX (TopPairing, Fact_Event_Results[Country_noc])

  VAR TopSport =
      MAXX (TopPairing, Fact_Event_Results[Sport])

  RETURN
      IF(ISBLANK (TopCountry),
      Blank(),
      TopCountry & " — " & TopSport)
```

Page 4 — Sport & Event Analysis

Business Insights

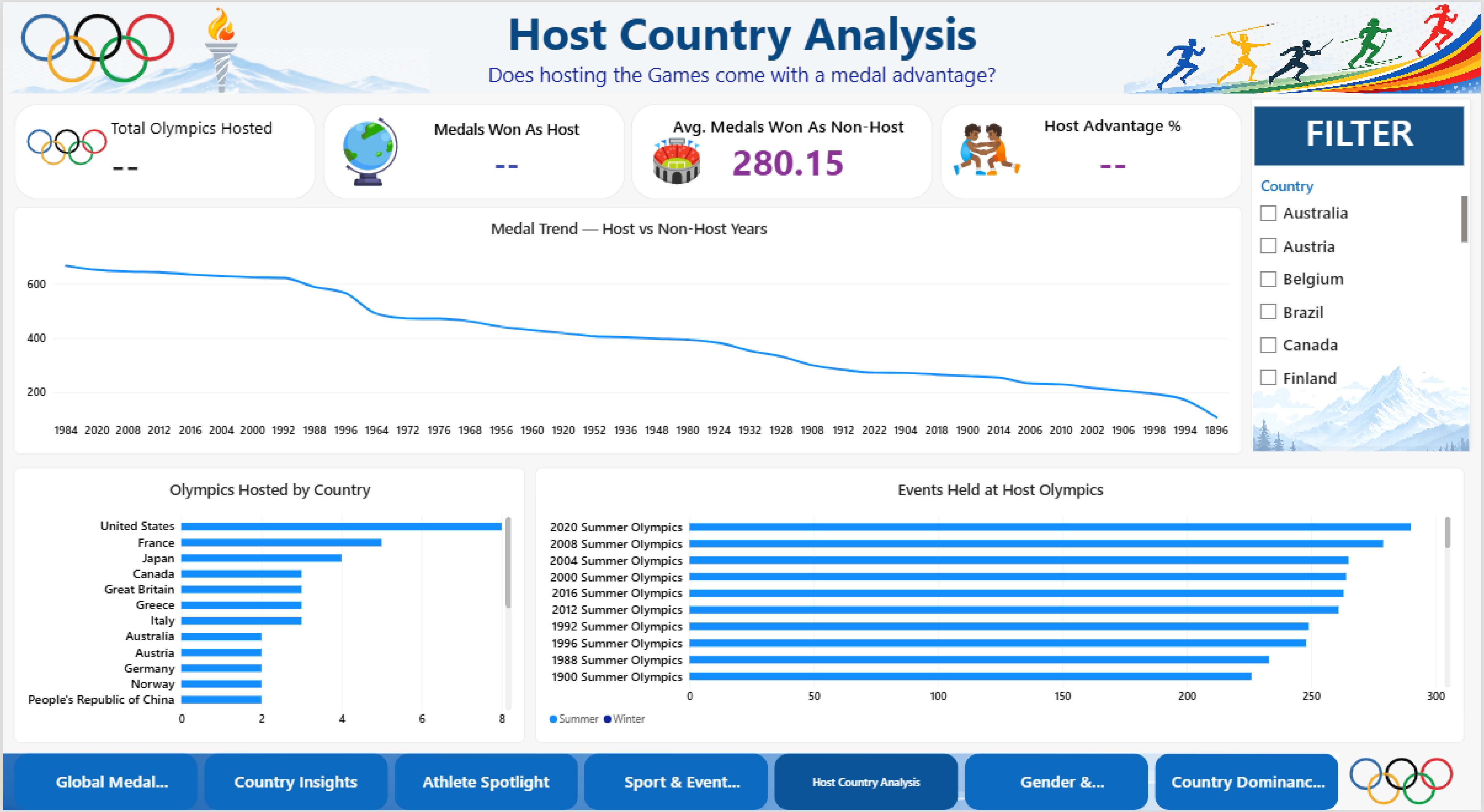
Total Sports	Total Events	Events Added	Most Dominant
112	963	+5	USA — Athletics

Business Insights

- USA–Athletics is the most dominant country–sport pairing with 4,500+ medal appearances. USA–Swimming follows at 3,300+. Dominance is even more concentrated at pairing level than at the country level alone.
- Athletics leads both total medals (4.5K) and athlete participation (30K+). Boxing ranks in the top 5 for participation but not for medals — confirming that the participation and medal-output charts tell different analytical stories.
- Sport-wise Medal Trends (Top 5): Athletics and Swimming show consistent growth from the 1900s. Artistic Gymnastics shows a sharp rise from the 1950s onwards.
- Events Added vs Previous Edition = +5. Season-matched comparison against the previous same-season Games ensures a meaningful, comparable result.
- Country Dominance by Sport confirms that the overall medal leader does not necessarily lead in every individual sport — dominance is discipline-specific.

Slicers: Season · Year Range: 1896–2022 · Sport (multi-select, 112 sports) · Drill-through to Country Dominance Detail

Page 5 — Host Country Analysis



Host Country Analysis

Does hosting the Games come with a medal advantage?

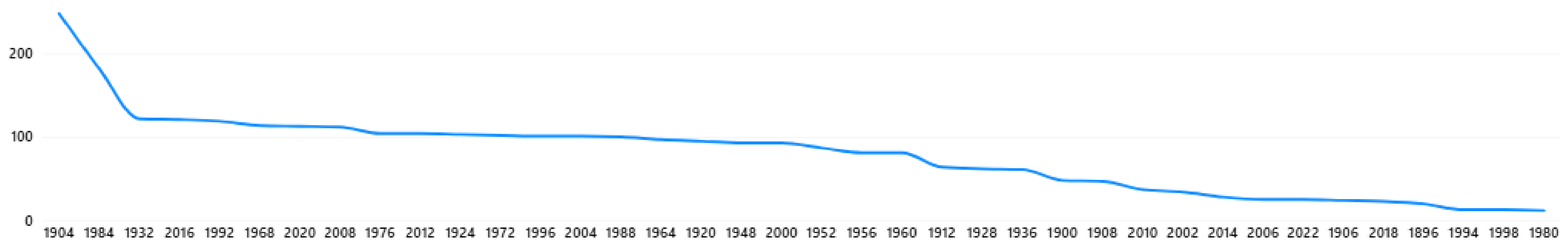
Total Olympics Hosted	Medals Won As Host	Avg. Medals Won As Non-Host	Host Advantage %
8	701	51.29	12.67

FILTER

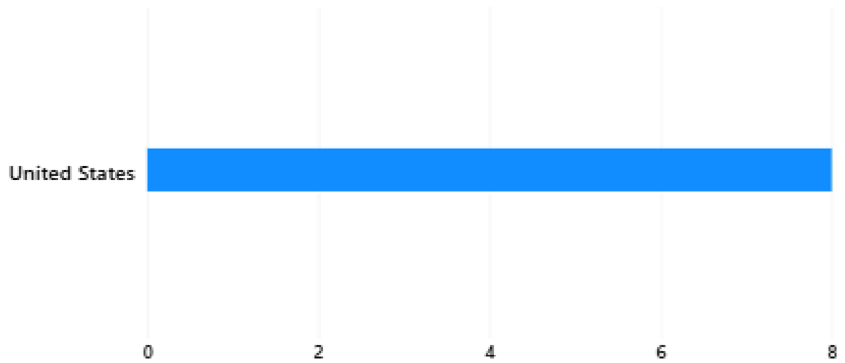
Country

- [ ] Spain
- [ ] Sweden
- [ ] Switzerland
- [x] United States
- [ ] West Germany
- [ ] Yugoslavia

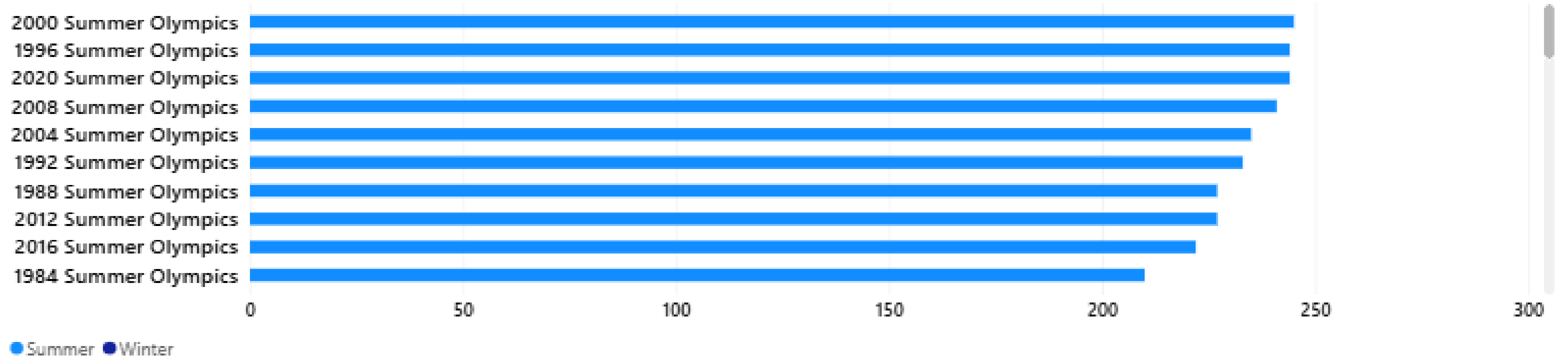
Medal Trend — Host vs Non-Host Years



Olympics Hosted by Country



Events Held at Host Olympics



Page 5 — Host Country Analysis

DAX Measures

DAX Measures & Analytical Logic — Page 5 — Host Country Analysis

Total Olympics Hosted

```
Total Olympics Hosted =
  VAR SelectedCountry =
      SELECTEDVALUE ( Dim_Country[country] )
  VAR SelectedNOC =
      CALCULATE(
            SELECTEDVALUE (Dim_Country[NOC]),
            FILTER (ALL(Dim_Country), Dim_Country[country] = SelectedCountry ) )
  RETURN
  CALCULATE (
      DISTINCTCOUNT ( Dim_Games[edition_id] ),
      FILTER (
            ALL ( Dim_Games ), Dim_Games[country_noc] = SelectedNOC
      && Dim_Games[Games_Status] = "Held" ) )
```

→ No Dim_Country → Dim_Games relationship in the model — FILTER(ALL()) bridges the lookup explicitly in DAX.

Medals Won As Host

```
Medals Won As Host =
  VAR SelectedNOC = SELECTEDVALUE ( Dim_Country[noc] )
  VAR HostEditions =
      FILTER ( ALL ( Dim_Games ),
      Dim_Games[country_noc] = SelectedNOC
      && Dim_Games[Games_Status] = "Held" )

  RETURN
  CALCULATE ( [Total Medals Awarded], HostEditions )
```

→ Calculates medals only for editions where the selected country was the host nation.

Avg. Medals Won As Non-Host

```
Avg. Medals Won As Non-Host =
  VAR HostEditions =
        CALCULATETABLE (
           VALUES ( Dim_Games[edition_id] ),
           FILTER ( Dim_Games, Dim_Games[country_noc] = SELECTEDVALUE(Dim_Country[noc] ) )
  RETURN
        CALCULATE (
           AVERAGEX ( VALUES ( Dim_Games[edition_id] ), [Total Medals Won] ),
           EXCEPT ( ALL ( Dim_Games[edition_id] ), HostEditions ),
           Dim_Games[Games_Status] = "Held" )
```

→ Average medals per non-host edition for the selected country — the comparison baseline for Host Advantage %.

Page 5 — Host Country Analysis

DAX Measures

DAX Measures & Analytical Logic — Page 5 — Host Country Analysis

Host Advantage %

```
Host Advantage % =
VAR SelectedNOC =
     CALCULATE (
          SELECTEDVALUE (Dim_Country[NOC]),
          FILTER (ALL (Dim_Country),
               Dim_Country[Country] = SELECTEDVALUE (Dim_Country[Country] ) ) )
RETURN
  IF (
     ISBLANK ( SelectedNOC), BLANK (),
    DIVIDE (
      [Medals Won As Host] - [Avg. Medals Won As Non-Host],
      [Avg. Medals Won As Non-Host], BLANK() ) )
```

→ ISBLANK guard returns blank when no country is selected — prevents misleading -100% from displaying as default.

→ Flags the 25 host nations on Dim_Country. Page 5 slicer is filtered to Is_Host_Country = "Yes" only.

Page 5 — Host Country Analysis

Business Insights

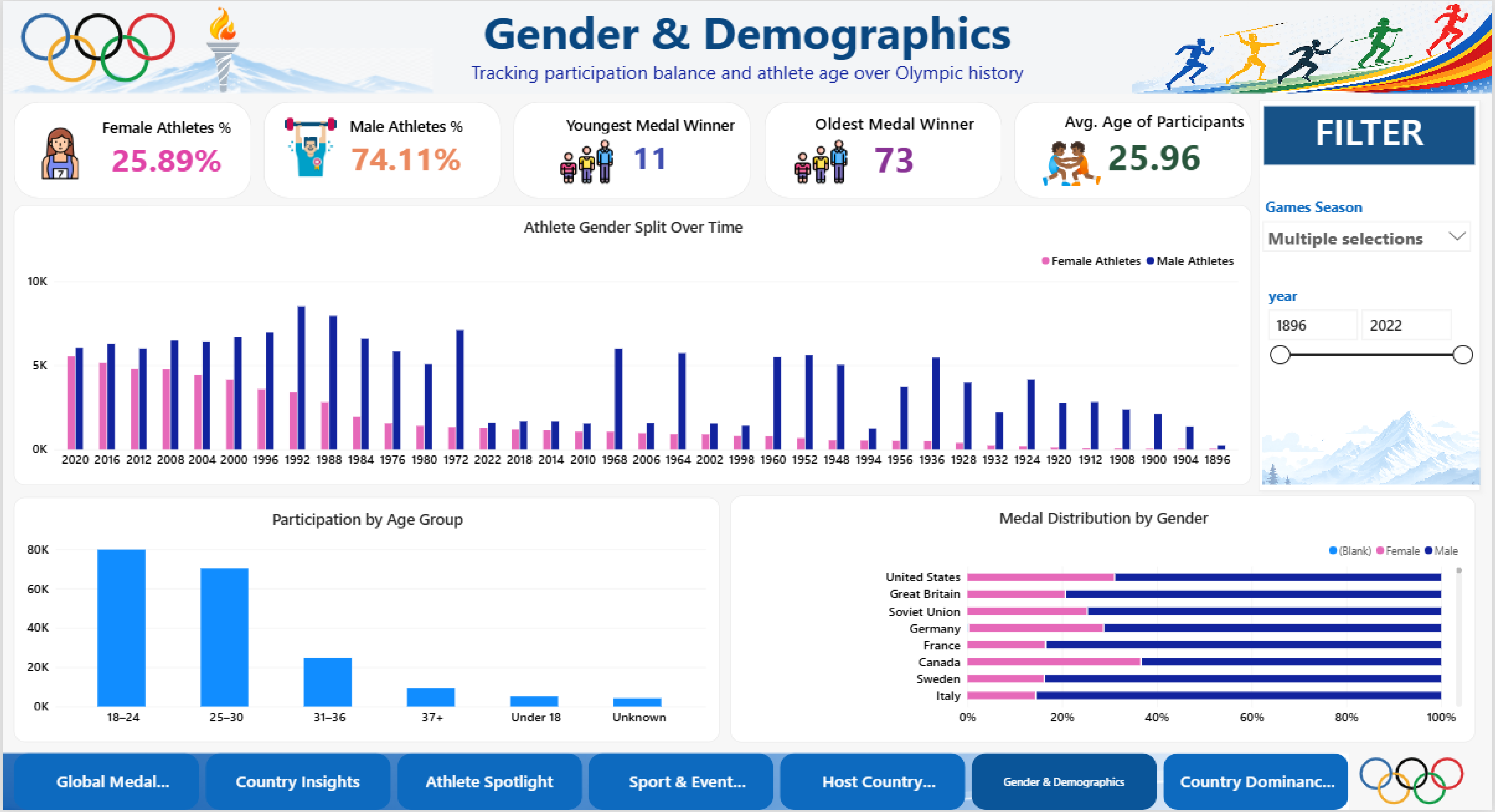
Host Nations	USA Hosted	Most Events	Slicer
25 of 234	8 Games	2020 Summer	Host nations only

Business Insights

- Only 25 of 234 countries have ever hosted the Olympic Games. The country slicer on this page is filtered to Is_Host_Country = "Yes" — preventing users from selecting non-host countries where a comparison is meaningless.
- USA has hosted 8 Olympic editions — the highest of any country, confirmed by the longest bar in the "Olympics Hosted by Country" chart on the dashboard.
- The 2020 Summer Olympics (Tokyo) leads the "Events Held at Host Olympics" chart with the most events of any single edition in the dataset.
- Host Advantage % is statistically meaningful only for multi-time hosts. For single-host nations, one result versus a historical average is directional evidence only — not a proven causal effect.
- When no country is selected, Host Advantage % returns blank — the card displays "Select a Country" as the default state, preventing a misleading -100% from appearing.

Slicers: Dim_Country[country] filtered to Is_Host_Country = "Yes" (25 nations only) · Season

Page 6 — Gender & Demographics



Page 6 — Gender & Demographics

DAX Measures

DAX Measures & Analytical Logic — Page 6 — Gender & Demographics

Male Athletes (counts)

```
Male Athletes =
  CALCULATE ( DISTINCTCOUNT ( Fact_Event_Results[athlete_id] ),
    Dim_Athlete[sex] = "Male"
```

Female Athletes (counts)

```
Female Athletes =
  CALCULATE ( DISTINCTCOUNT ( Fact_Event_Results[athlete_id] ),
    Dim_Athlete[sex] = "Female"
```

Male Athletes (%)

```
Male Athletes % =DIVIDE ( [Male Athletes], [Male Athletes] + [Female Athletes], BLANK() )
```

Female Athletes (%)

```
Female Athletes % = DIVIDE ( [Female Athletes], [Male Athletes] + [Female Athletes], BLANK() )
```

Youngest Medal Winners

```
Youngest Medal Winner =
  CALCULATE (
    MIN ( Fact_Event_Results[Age_At_Edition] ),
    Fact_Event_Results[medal] <> BLANK() )
```

Oldest Medal Winners

```
Oldest Medal Winner Age =
  CALCULATE (
    MAX ( Fact_Event_Results[Age_At_Edition] ),
    Fact_Event_Results[medal] <> BLANK() )
```

Page 6 — Gender & Demographic [redacted]

DAX Measures

DAX Measures & Analytical Logic — Page 6 — Gender & Demographics

Age_Band & Age_Band_Sort (Power Query)

```
Age_Band (Power Query custom column):
  if [Age_At_Edition] < 18 then "Under 18"
  else if [Age_At_Edition] <= 24 then "18-24"
  else if [Age_At_Edition] <= 30 then "25-30"
  else if [Age_At_Edition] <= 36 then "31-36"
  else "37+"
```

→ Created in Power Query — not as DAX calculated columns. A DAX column cannot sort another DAX column in the same table (causes circular dependency error).

Gender Distribution of Medal Winners

```
Gender Distribution of Medal Winners =
CALCULATE (
      DISTINCTCOUNT (Fact_Event_Result[athelete_id]),
      Fact_Event_Result[medal] <> BLANK ()
)
```

Avg. Age of Participants

```
Avg. Age of Participants = AVERAGE ( Fact_Event_Results[Age_At_Edition] )
```

→ Average age across all participating athletes (no medal filter) — reflects the full competitor age profile.

Page 6 — Gender & Demographics

Business Insights

Female Athletes %	Male Athletes %	Youngest Winner	Oldest Winner	Avg. Age
25.89%	74.11%	11 yrs	73 yrs	25.96 yrs

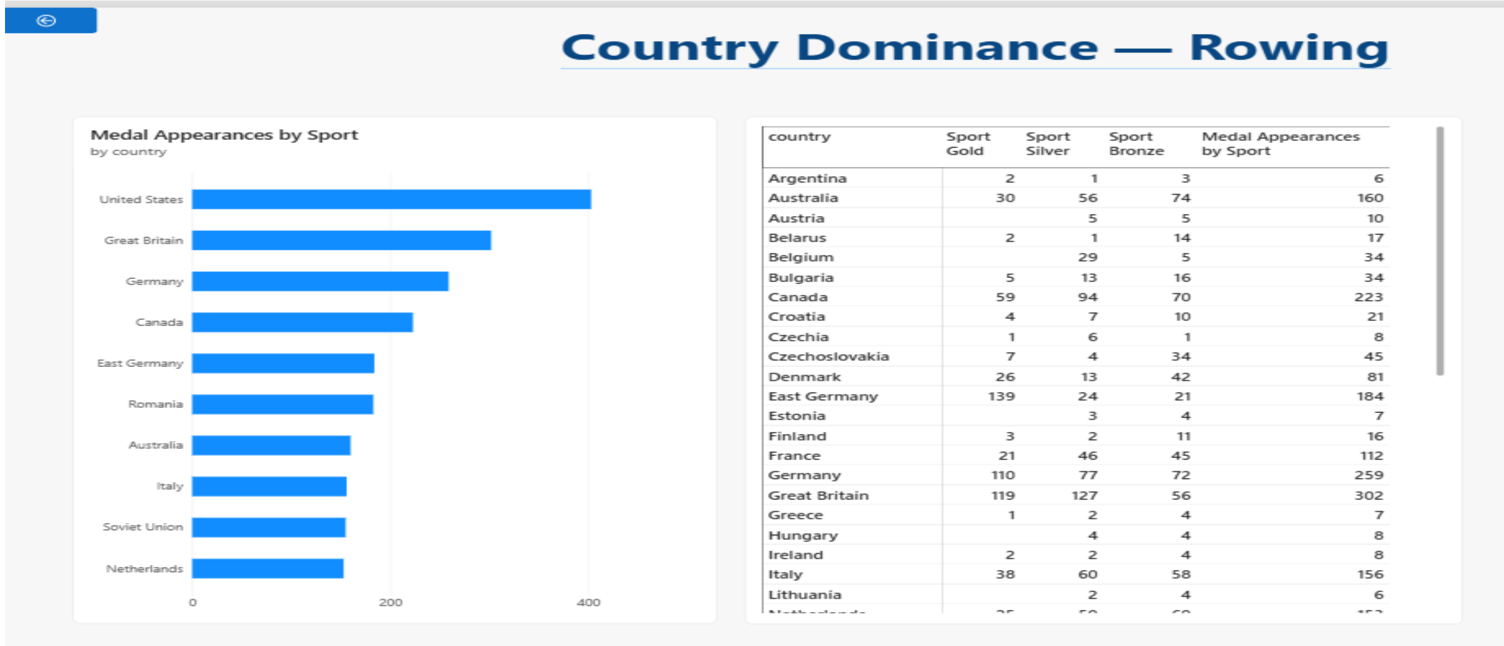
Business Insights

- Female athletes represented 0% at the 1896 Athens Games and have grown to 25.89% across all-time records — a sustained, visible increase across 126 years shown in the Gender Split Over Time chart.
- The 25–30 age band is the largest group at medal wins, followed by 18–24. The Oldest Medal Winner at 73 comes from a sport with no effective upper age restriction — age demographics are fundamentally sport-dependent.
- Male medal winners dominate the Medal Distribution by Gender chart for top nations — reflecting both the longer history of male participation and the historic imbalance in available events across 1896–2022.
- The gender split at medal wins (Male 72.12%, Female 27.87%) differs from the overall participation split, indicating that female athletes have a proportionally different medal-win rate across different sports and eras.

Slicers: Season (Summer / Winter) · Year Range: 1896–2022

Drill-Through Pages

Two interactive detail pages triggered by right-clicking a visual on the source page



country	Sport Gold	Sport Silver	Sport Bronze	Medal Appearances by Sport
Argentina	2	1	3	6
Australia	30	56	74	160
Austria		5	5	10
Belarus	2	1	14	17
Belgium		29	5	34
Bulgaria	5	13	16	34
Canada	59	94	70	223
Croatia	4	7	10	21
Czechia	1	6	1	8
Czechoslovakia	7	4	34	45
Denmark	26	13	42	81
East Germany	139	24	21	184
Estonia		3	4	7
Finland	3	2	11	16
France	21	46	45	112
Germany	110	77	72	259
Great Britain	119	127	56	302
Greece	1	2	4	7
Hungary		4	4	8
Ireland	2	2	4	8
Italy	38	60	58	156
Lithuania		2	4	6

Country Dominance — Rowing (drill-through from Page 4)

DAX — Sport-Specific Measures (used in drill-through table)

```
Sport Gold   = CALCULATE ( COUNTROWS ( Fact_Event_Results ),
Fact_Event_Results[medal] = "Gold" )
Sport Silver = CALCULATE ( COUNTROWS ( Fact_Event_Results ),
Fact_Event_Results[medal] = "Silver" )
Sport Bronze = CALCULATE ( COUNTROWS ( Fact_Event_Results ),
Fact_Event_Results[medal] = "Bronze" )
```

A. Country Insights Drill-Through

Triggered from: Page 1 — Top 10 Countries bar chart
Drill field: Dim_Country[country] · Keep all filters: ON

Delivers the full Page 2 Country Insights view pre-filtered to the right-clicked country — athletes, medals, efficiency, sport breakdown, and year trend — in one click. No manual re-selection required.

B. Country Dominance Detail

Triggered from: Page 4 — Sport & Event charts
Drill field: Fact_Event_Results[sport]
Dynamic title: "Country Dominance — [Selected Sport]"

Visuals: Bar chart (Top 10 countries in sport) + Table with Sport Gold, Sport Silver, Sport Bronze, Medal Appearances by Sport

Sport Gold/Silver/Bronze source from Fact_Event_Results — not Fact_Medal_Tally. This ensures values are correctly scoped to the selected sport only.

Challenges Faced & Key Learnings

Six real issues identified and resolved during the build

Team-Sport Double-Count

Fact_Event_Results has one row per athlete per team medal. A 12-player gold-medal team produces 12 rows there — not 1. All country-level medal totals exclusively use Fact_Medal_Tally (1 row per country per edition).

edition_id vs edition Text Keys

Two editions had different text spellings across tables, orphaning 342 rows and injecting a phantom blank row into Dim_Games — causing Edition_Sequence to start from 2. Fixed: all Dim_Games relationships switched to edition_id.

Auto Date/Time — Cyclic Reference

Power BI Auto date/time built hidden LocalDateTable objects that closed a relationship loop, blocking all refreshes. Fixed by disabling under File → Options → Current File → Data Load.

Age Calculation Producing Values > 1,000

Non-standard birth date text formats produced malformed 3-digit years. Fixed: Born_Year rebuilt in Power Query with digit extraction and a 1850–present range check. Age_At_Edition bounded 10–75.

Context Transition in MAXX

Most Olympic Appearances initially returned 24 for every athlete (the global edition count). DISTINCTCOUNT without inner CALCULATE ignores iterator context. Fixed: CALCULATE(DISTINCTCOUNT(edition_id)) inside MAXX.

Events Added Returning −196

Compared 2022 Winter (94 events) to 2020 Summer (290 events) — a cross-season comparison. Fixed: season-matching added before selecting the prior edition. Dashboard now correctly shows +5.

Business Recommendations

Six evidence-based recommendations from the actual dataset

1 Use Medal Efficiency, not raw totals

Raw counts favour large, long-tenured programmes structurally. Efficiency (medals per 100 athletes) is the correct normalised metric for cross-country programme evaluation and federation ROI decisions.

2 Host advantage — per country, not universal

Multi-time hosts show visible medal spikes. For single-host nations, one result vs. a historical average is directional only. Bid strategy discussions should reference per-country data, not a blanket assumption.

3 Focus on 2–5 sports, not breadth

Norway, China, Sweden, and Japan all show clear sport-level concentration in their top disciplines. Targeted investment outperforms broad multi-sport strategies on medal efficiency.

4 Distinguish participation from medal production

Athletics and Swimming lead on both metrics. Other sports diverge — high participation does not guarantee proportional medal returns. Use the metric relevant to the specific investment objective.

5 Gender parity requires era-based monitoring

The 74:26 all-time ratio reflects 126 years of data skewed by early male-only decades. Era-based trend analysis is more actionable for IOC policy decisions than the all-time aggregate figure.

6 Data ends at 2022 — state this boundary explicitly

Paris 2024 and subsequent Games are not in this dataset. Any recommendation referencing "recent performance" must clearly state the data boundary to avoid misrepresentation.

Thank You

Olympic Insights Using Data Analytics

- 6 CSV source files profiled, cleaned, and modeled into a validated star schema
- 315,626 event records across 55 held Olympic editions — 1896 to 2022
- 7 validated relationships · all edge cases confirmed against live model data
- All KPIs tested through visual cross-checks — corrected before submission
- 6 dashboard pages + 2 drill-through pages covering all assignment requirements

Presented by Kavita Kanwar Naruka

Power BI · Power Query · DAX · Star Schema